DATAOPS SOC | NOC | VAPT INNOVATION HACKATHON

AI Agent for Automated Penetration Testing

The Complete Beginner-to-Architect Reference Manual

A comprehensive, line-by-line master guide designed for complete beginners, developers, and senior evaluators. Explaining every cybersecurity concept, tool mechanism, algorithmic engine, mathematical scoring formula, and presentation defense from scratch.

WHAT THIS MASTER REFERENCE MANUAL EMPOWERS YOU TO DO:

- Understand all cybersecurity fundamentals using simple, everyday physical analogies (houses, locks, doctors).
- Master the exact Problem Statement from Page 1 of the hackathon brief and explain why manual VAPT fails.
- Master all 7 required security tools: Nmap, Nikto, OWASP ZAP, Nessus, OpenVAS, Burp Suite, and Metasploit.
- Learn how this software platform was engineered from zero using React 19, TypeScript, Express, and Vite.
- Understand the Five Core Algorithmic Engines: Scope Enforcement, AI Planning, Tool Orchestrator, Cross-Tool Correlation, and Human-in-the-Loop Gating.
- Master industry standards: NIST SP 800-115, OWASP Top 10, CVE, CVSS v3.1 math, CWE, and MITRE ATT&CK.
- Learn how to present, pitch, and defend this solution before hackathon judges, technical architects, and clients.
- Rebuild and code this entire platform with your own hands without any external assistance.

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Foreword: Why We Wrote This Guide

Cybersecurity is frequently treated like an arcane branch of black magic, shrouded in impenetrable jargon, acronyms (CVSS, CVE, CWE, DAST, BOLA, IDOR), and fragmented command-line flags. Newcomers and developers often feel intimidated, while executive decision-makers struggle to understand why their technical teams are overwhelmed.

The purpose of this handbook is to break down every single wall. Whether you have never typed a command into a terminal in your life, or you are an experienced software engineer preparing to defend a multi-tiered architecture in front of demanding hackathon judges, this book will guide you step-by-step.

The Three Pillars of This Guide:

- **1. Real-World Physical Analogies:** Every complex cybersecurity mechanism (ports, packets, injection, path traversal) is paired with a concrete, physical real-world scenario (houses, revolving doors, bank tellers, customs officers). You will never be asked to memorize an acronym without understanding its physical meaning.
- **2. Concrete Technical Rigor:** Nothing is hidden behind "magic." You will see the exact network packets sent over the wire, the raw HTTP requests and responses, the full terminal outputs, and the TypeScript code snippets that parse and normalize them.
- **3. Presentation & Defense Readiness:** This handbook contains the exact 15-minute presentation script, slide-by-slide choreography, and 10 tough judge defense scenarios so you can present this platform with supreme authority.

Ø=Ü¡ HOW TO NAVIGATE THIS HANDBOOK FOR YOUR ROLE

- If you want to understand cybersecurity from scratch: Read Chapters 1, 2, and 16.
- If you want to master the 7 required tools: Read Chapters 3 through 9.
- If you want to understand the software architecture and algorithms: Read Chapters 10, 11, and 15.
- If you are presenting or defending this project in front of judges: Read Chapters 2, 12, 13, and 14.

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CHAPTER 01

Cybersecurity 101: The Fortress & Inspector Analogy

Mastering the core principles of computer security with zero technical prerequisites

1.1 What is Cybersecurity in Everyday Language?

Imagine you own a large, high-tech commercial building located in a bustling metropolitan city. Inside the building are several sensitive areas: on the ground floor is a public customer lobby; on the second floor are corporate offices with confidential files; in the basement is a high-security bank vault holding millions of dollars in cash and bearer bonds; and on the roof is a private helipad with communication antennae.

Every day, hundreds of legitimate people enter and exit: bank customers visit the lobby, office employees badge in to work, delivery trucks unload supplies at the loading dock, and cleaners arrive in the evening. At the same time, thousands of potential criminals, opportunistic thieves, and industrial spies are walking down the street outside.

Cybersecurity is simply the discipline of designing, inspecting, and defending this building so that authorized visitors can enter easily to conduct business, while unauthorized burglars, trespassers, and vandals are prevented from entering, stealing data, or causing physical destruction.

Ø=Ü¡ THE GOLDEN ANALOGY: VULNERABILITY VS. EXPLOIT VS. THREAT VS. RISK

To understand cybersecurity, you must understand four foundational terms that are often confused:

1. VULNERABILITY (The Weakness): A flaw or defect in the building. Example: A second-floor window has a rusted latch that doesn't lock properly.

2. THREAT (The Danger): An outside force capable of doing harm. Example: A professional burglar walking down the alley with a flashlight and crowbar.

3. EXPLOIT (The Action/Tool): The exact method used to take advantage of the weakness. Example: The burglar using the crowbar to pry open that rusted window latch.

4. RISK (The Consequence): How likely is it that the burglar will find the window, and what can they steal once inside? If the window opens into an empty broom closet, risk is LOW. If it opens directly into the bank vault, risk is CRITICAL.

1.2 What is an IP Address, a Port, and a Packet?

When you look at a digital network, you often hear terms like "IP Address", "Port Number", and "Packet". Here is how to understand them without any networking degree:

- **IP Address = The Street Address:** An IP Address (like 10.0.12.45 or 192.168.1.1) is the exact physical street address of the building. Just like postal delivery trucks cannot deliver mail without knowing the street name and building number, computers cannot deliver digital messages without knowing the target IP address.
- **Port = The Numbered Entrance Door:** A single computer server can run dozens of different software programs simultaneously (a website, an email server, a database, a remote management console). A Port is like a specific numbered entrance door or window on that building. There are 65,535 possible port numbers on every computer!

- **Port 80 (HTTP):** Port 80 is the standard revolving front door for public web pages (HTTP). Anyone on the street is welcomed in.
- **Port 443 (HTTPS):** Port 443 is the armored front entrance where visitors must show identification and all communications are encrypted so spies cannot eavesdrop (HTTPS).
- **Port 22 (SSH):** Port 22 is the heavy steel maintenance door behind the building used by administrators to log in remotely (SSH - Secure Shell).
- **Port 21 (FTP):** Port 21 is the loading dock used for shipping and receiving cargo files (FTP - File Transfer Protocol).
- **Port 5432 (Database):** Port 5432 or 3306 is the subterranean vault door where structured database records are kept (PostgreSQL / MySQL).
- **Packet = The Sealed Envelope:** A Packet is like a sealed envelope containing a letter. When computers talk, they chop their data into small chunks called packets, stamp the destination IP and Port on the outside, and send them across the internet highway.

1.3 The CIA Triad: The Three Cornerstones of Security

Every security vulnerability in the world is measured by how it threatens one or more pillars of the "CIA Triad". These pillars are not related to the Central Intelligence Agency—they stand for Confidentiality, Integrity, and Availability:

• **1. Confidentiality (Secrecy):** Only authorized people can see sensitive information. In our building, only bank executives can open customer account drawers. If a hacker steals customer credit card numbers, Confidentiality is broken.

• **2. Integrity (Accuracy & Trust):** Information must remain accurate and cannot be tampered with by unauthorized parties. In our building, no customer can erase their mortgage balance or write extra zeroes on their checking balance. If a hacker alters banking records, Integrity is broken.

• **3. Availability (Uptime & Access):** The building and services must be accessible whenever authorized people need them. If a vandal superglues all the door locks shut or floods the lobby with smoke so no customer can enter, Availability is broken (this is what a Denial-of-Service attack does).

1.4 What is Penetration Testing (VAPT)?

VAPT stands for Vulnerability Assessment and Penetration Testing. It is a legal, authorized, controlled security evaluation where a certified ethical security analyst (a "white-hat hacker") pretends to be an attacker.

Imagine hiring an undercover security auditing firm to test your bank. You give them written legal authorization (called Rules of Engagement) to walk around your building at night, rattle all the doorknobs, check if the security cameras have blind spots, test if the backdoor lock can be picked, and see if the receptionist can be tricked into handing over a master key.

However, unlike real criminals, the ethical tester NEVER steals money, NEVER breaks the furniture, and NEVER harms the building. When the audit is finished, they write a comprehensive VAPT Report that lists every broken lock discovered, details how dangerous each flaw is, and gives step-by-step instructions on how to repair each one before real thieves arrive.

&–þ THE LEGAL BOUNDARY: WHITE HAT VS BLACK HAT

The difference between a 10-year prison sentence and a $200,000 security consulting fee is ONE piece of paper: The Rules of Engagement (RoE).

• A White-Hat Hacker has explicit, written legal authorization from the asset owner to test specific IP addresses during designated time windows, with clear safety constraints.

• A Black-Hat Hacker probes networks without authorization, intending to steal data, extort ransoms, or cause destruction.

In our platform, the Scope Enforcement Engine acts as a legal firewall: if an analyst accidentally enters a domain like "google.com", the platform immediately blocks the scan with a Critical Scope Violation alert!

CHAPTER 02

Deconstructing Problem Statement Page 1 Line-by-Line

Why automated VAPT agents are revolutionary and what the hackathon demanded

2.1 The Core Background: Why Traditional Penetration Testing Fails

On Page 1 of the official hackathon document, Section 1 states:

"Penetration testing involves several stages, including reconnaissance, attack-surface discovery, scanning, enumeration, vulnerability analysis, validation, and reporting. These stages typically require multiple tools and substantial manual correlation. Cybersecurity analysts may spend more time combining and interpreting outputs from different tools than actually analyzing security risks."

Let us break down why this is such a massive problem in the modern world:

- **The 7 Stages of Testing:** Stage 1: Reconnaissance (Scouting the target from afar).

Stage 2: Attack-Surface Discovery (Identifying all public servers, IPs, and domain names).

Stage 3: Scanning (Probing all ports to see which doors are open).

Stage 4: Enumeration (Querying each open service to find its exact software version).

Stage 5: Vulnerability Analysis (Comparing versions against known security flaws).

Stage 6: Validation (Verifying whether the flaw is real or a false alarm).

Stage 7: Reporting (Writing the client report with executive summary and remediation roadmap).

Ø=Ü¡ THE MEDICAL CLINIC ANALOGY FOR CYBERSECURITY TOOL FATIGUE

Imagine a large hospital clinic with 10 diagnostic machines: Machine 1 measures heart rate; Machine 2 measures blood sugar; Machine 3 scans lungs; Machine 4 checks brain waves. None of these machines connect to a common computer! Every machine prints a separate 50-page paper report using different abbreviations and formats.

The head doctor is forced to spend 8 hours every day sitting at a desk with highlighters, manually copying data from 10 paper printouts into a single Word document, manually crossing out duplicate readings, and trying to spot patterns! By the time they finish the paperwork, they are exhausted and have almost no time left to actually treat the patient!

This is the daily reality of cybersecurity analysts. They spend 80% of their billable hours fighting tool noise, formatting CSVs, and deduplicating alerts rather than securing networks.

2.2 The Six Mandatory Capabilities Explained Simply

Page 1, Section 2 of the problem statement explicitly lists six core capabilities that the AI Agent must demonstrate. Here is what each one means in plain language:

- **1. Selecting Appropriate Tools:** A doctor does not order a brain MRI if the patient has a sprained ankle. Similarly, the agent must not run database scanners against an FTP server. It must pick the exact right tool for the discovered service.
- **2. Interpreting Scan Results:** Raw tool outputs look like thousands of lines of unformatted text or dense XML. The agent must read this telemetry, extract the key findings, and understand their technical meaning.
- **3. Deciding What Needs to Be Investigated Next:** Security testing is an adaptive investigation. When the agent finds Apache 2.4.49 on Port 80, it must immediately recognize that this version has a famous directory traversal flaw (CVE-2021-41773) and decide to inspect web directories next.
- **4. Correlating Evidence from Multiple Tools:** If Nikto, Nessus, and OpenVAS all find the same Apache bug on Port 80, the agent must not list it three times! It must correlate the evidence, merge them into a single finding, and keep all evidence links intact.
- **5. Prioritizing Identified Risks:** The agent must mathematically rank vulnerabilities from Critical down to Low based on CVSS 3.1, exposure, and exploitability—without fabricating fake CVE numbers or scores.
- **6. Explaining Decisions & Reasoning:** The agent must provide an auditable decision trace explaining WHY it performed each action (e.g. "Dispatched Nikto because Port 80 was discovered open with Apache httpd").

2.3 Reasoning vs Scripted Execution (The Winning Factor)

The brief explicitly emphasizes: "The agent must demonstrate actual security reasoning rather than blindly executing a predefined sequence of scanners."

Anyone can write a simple bash script that runs 7 tools in a fixed pipeline. But a script is blind: it runs Tool 3 even if Tool 1 found that the host is completely offline! In contrast, our AI Agent functions like a senior security consultant:

- **Adaptive Branching:** If Port 80 is closed, it cancels all web scanners (Nikto, ZAP, Burp) and conserves bandwidth.
- **Contextual Pivoting:** If it discovers an unauthenticated API endpoint, it automatically schedules Burp Suite and OWASP ZAP to test for BOLA and SQL Injection.
- **Evidence Grounding:** It distinguishes raw telemetry from AI inference, ensuring that every finding is grounded in physical evidence.

CHAPTER 03

Tool Deep-Dive 1: Nmap (Network Mapper)

The undisputed gold standard for network reconnaissance and port discovery

3.1 What is Nmap in Everyday Terms?

Nmap (Network Mapper) was created in 1997 by Gordon Lyon (Fyodor). It is the very first tool every security analyst, network administrator, and ethical hacker runs when assessing a target.

In our fortress analogy, Nmap is the reconnaissance scout who approaches the building from the street with a clipboard. The scout walks around the perimeter, taps on every numbered door and window, checks whether the door is open or locked, and writes down the exact brand and model of every lock installed.

3.2 How Nmap Works Under the Hood (TCP SYN Packets)

How can a computer program know if a port is open across the world? Nmap uses the fundamental rules of the TCP/IP protocol (the 3-Way Handshake).

- **The Normal TCP Handshake:** When computer A wants to talk to computer B, it sends a SYN (Synchronize) packet. If computer B is listening, it replies with a SYN/ACK (Synchronize/Acknowledge). Computer A then sends an ACK, and communication begins.
- **Nmap Stealth SYN Scan (-sS):** Nmap sends a SYN packet. If the remote port is OPEN, the target sends back a SYN/ACK. But instead of completing the connection, Nmap immediately sends an RST (Reset) packet to tear down the connection! This is called a "Half-Open SYN Stealth Scan". It reveals that the port is open without creating a full application connection, minimizing log noise on the target.
- **Service Version Detection (-sV):** Once a port is known to be open, Nmap sends specific application probe messages to see how the service responds. An Apache web server responds with "Server: Apache/2.4.49 (Unix)". An SSH server responds with "SSH-2.0-OpenSSH_7.4p1". This is called Banner Grabbing or Version Detection (-sV).

3.3 Key Command Flags Used in Our Platform

NMAP COMMAND SYNTAX

```
# Command executed against authorized lab sandbox:
nmap -sS -sV -sC -Pn -T4 -p 21,22,80,443,5432,8080 10.0.12.45

# Flag Breakdown:
# -sS : TCP SYN Stealth Scan (half-open, fast, stealthy)
# -sV : Probe open ports to determine service/version info
# -sC : Run default NSE scripts (checks for anonymous FTP, SSL ciphers)
# -Pn : Skip ping host discovery (treat target as online)
# -T4 : Aggressive timing template (optimizes speed for lab testing)
# -p  : Explicit port list to scan
```

3.4 Raw Output Discovered in Our Platform

Here is the actual genuine output discovered on 10.0.12.45:

NMAP CONSOLE TRANSCRIPT

```
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4 (Anonymous login permitted)
22/tcp   open  ssh         OpenSSH 7.4p1 (protocol 2.0)
80/tcp   open  http        Apache httpd 2.4.49 ((Unix))
443/tcp  open  ssl/http    Apache httpd 2.4.49 ((Unix)) (TLS 1.0/1.1 enabled)
5432/tcp open  postgresql  PostgreSQL DB 14.2
8080/tcp open  http        Apache Tomcat / Spring Boot 2.6.3
```

3.5 How the Agent Normalized Nmap into Schema JSON

The Tool Ingestion Engine converts the raw Nmap text into our standardized Common Finding Schema. Notice how the evidence block preserves the exact command executed:

NORMALIZED NMAP FINDING

```
{
  "id": "NMAP-001",
  "tool": "Nmap",
  "affectedAsset": "10.0.12.45",
  "port": 443,
  "service": "ssl/http",
  "title": "Deprecated TLS 1.0 and 1.1 Cryptographic Protocol Enabled",
  "severity": "Medium",
  "cvssScore": 5.3,
  "cweId": "CWE-326",
  "confidence": "Confirmed",
  "commandExecuted": "nmap -sS -sV -sC -Pn -p 443 10.0.12.45"
}
```

CHAPTER 04

Tool Deep-Dive 2: Nikto Web Scanner

The specialized web server configuration and known-issue auditor

4.1 What is Nikto in Everyday Terms?

Nikto is an open-source web server scanner written by Chris Sullo. While Nmap checks general network ports, Nikto specializes specifically in web servers running on Ports 80 and 443.

In our fortress analogy, Nikto is the municipal building inspector who enters the front lobby and inspects the building codes. The inspector checks whether the emergency exit signs are lit, whether hazardous chemicals are stored in public hallways, whether old employee files were left in unlocked cardboard boxes, and whether the front door hinges are rusted.

4.2 How Nikto Works Under the Hood

Nikto maintains a comprehensive database of over 6,700 dangerous files, outdated program versions, and web server configuration vulnerabilities. It rapidly fires HTTP GET, HEAD, and OPTIONS requests against the web server:

- **Exposed Source Code Repositories:** Nikto checks if developers accidentally left Git version control folders in the public web directory. In our lab target, Nikto discovered "/.git/HEAD" returning HTTP 200 OK! This means an attacker can download the entire git history and reconstruct the application's private source code!
- **Security Header Audit:** Nikto inspects HTTP response headers. It checks if anti-clickjacking headers (X-Frame-Options) or content-sniffing protections (X-Content-Type-Options) are missing.
- **Path Traversal Probing (CVE-2021-41773):** When Nikto saw Apache 2.4.49 on Port 80, it fired a test path traversal pattern: "GET /icons/.%%32%65/.%%32%65/etc/passwd". The server responded with HTTP 200 OK and leaked the Unix user accounts! This confirmed CVE-2021-41773 with zero doubt.

4.3 Command Flags & Raw Output Capture

NIKTO EXECUTION TRANSCRIPT

```
# Nikto execution command:
nikto -host 10.0.12.45 -port 80 -Format json -output /output/nikto.json

# Raw Output Discovered:
- Server: Apache/2.4.49 (Unix)
- Root page /: Anti-clickjacking X-Frame-Options header is not present.
- /icons/.%%32%65/.%%32%65/etc/passwd: Apache 2.4.49 Path Traversal flaw (CVE-2021-41773).
Confirmed: File retrieved with HTTP status 200.
- /.git/HEAD: Git repository directory found. Allows reconstruction of source code repository.
```

4.4 Significance of Nikto Findings in Our Assessment

Nikto proved crucial for identifying configuration hygiene flaws that general network scanners miss. The discovery of the exposed `.git` directory represents a massive risk: using tools like `git-dumper`, an adversary can reconstruct the entire code repository, discover API keys, database connection strings, and hidden architectural blueprints.

CHAPTER 05

Tool Deep-Dive 3: OWASP ZAP (Zed Attack Proxy)

Dynamic Application Security Testing (DAST) and active injection fuzzing

5.1 What is OWASP ZAP in Everyday Terms?

OWASP ZAP (Zed Attack Proxy) is one of the world's most popular free, open-source web application security testing tools, maintained by the Open Worldwide Application Security Project.

In our fortress analogy, ZAP is an undercover security auditor testing an automated bank teller machine (ATM). The auditor types random symbols into the PIN pad, tries entering letters where numbers are expected, attempts to withdraw negative amounts of money, and inspects whether the machine malfunctions or gives away free cash!

5.2 How ZAP Discovered the Critical SQL Injection (CWE-89)

On Port 8080 of our target, Nmap and Nikto discovered a Spring Boot REST API. The AI Agent adaptively decided to dispatch OWASP ZAP to fuzz the API parameters.

When ZAP spidered the banking application, it discovered an endpoint: /api/v1/transfer?account_id=1045. ZAP then performed Active Scanning: it injected SQL syntax characters like single quotes ('), double quotes ("), OR 1=1, and UNION SELECT.

Ø=Ü¡ WHAT IS SQL INJECTION (SQLi) IN PLAIN ENGLISH?

Imagine a drive-through banking tube where you send a paper slip to the teller. The slip says: "Please look up account number: [_____]".

A regular customer writes "1045", and the teller fetches account 1045. But a malicious hacker writes: "1045 OR 1=1; DROP TABLE accounts;". If the teller blindly obeys whatever text is written on the slip without verifying that it is just a plain number, the database executes the malicious command! In our lab, the injected payload caused the database to return all customer bank accounts and balances!

5.3 Raw HTTP Request & Response Proof

OWASP ZAP SQLI EVIDENCE PROOF

```
# OWASP ZAP Active Scan Proof Captured in Platform:
Target Endpoint: http://10.0.12.45:8080/api/v1/transfer?account_id=
Injected Attack Payload: account_id=1045%27%20OR%201=1--

HTTP Request:
GET /api/v1/transfer?account_id=1045%27%20OR%201=1-- HTTP/1.1
Host: 10.0.12.45:8080
Accept: application/json

HTTP Response (200 OK):
{
  "status": "success",
  "records": [
    { "id": 1001, "name": "admin", "balance": 994020.50 },
    { "id": 1002, "name": "corporate_treasury", "balance": 4850230.12 }
  ]
}
```

5.4 Remediation for SQL Injection

The permanent fix for SQL Injection is Parameterized Queries (Prepared Statements) or Object-Relational Mapping (ORM) frameworks. When queries are parameterized, user input is treated strictly as literal data, never as executable SQL instructions, rendering injection impossible.

CHAPTER 06

Tool Deep-Dive 4: Tenable Nessus

Enterprise infrastructure vulnerability auditing and native XML result ingestion

6.1 What is Nessus in Everyday Terms?

Tenable Nessus is the most widely deployed enterprise vulnerability assessment platform in the Fortune 500. It contains over 150,000 security plugins.

In our fortress analogy, Nessus is a certified team of master civil engineers carrying massive binders from the municipal safety board. The engineers walk through the building, cross-referencing every model of elevator, air filter, and fire door against a global recall list.

6.2 Plugin 153723 & Licensing Transparency

In our platform, Nessus was integrated via native .nessus XML file import. Nessus Plugin ID 153723 flagged Apache 2.4.49 on Port 80:

- **Plugin 153723:** Nessus Plugin 153723 checks for Apache HTTP Server 2.4.49 path traversal and file disclosure.
- **Plugin 112108:** Plugin ID 112108 flagged unauthenticated Spring Boot Actuator endpoints on Port 8080 (/actuator/env, /actuator/heapdump), warning that attackers could download server memory containing database passwords.
- **Licensing Transparency:** Under Section 6 of the hackathon brief: "Teams must clearly disclose which integrations execute live scans and which only import recorded outputs. Commercial tool availability and licensing must not be misrepresented." Because Nessus is commercial proprietary software, our matrix truthfully classifies it as IMPORT-ONLY, ingesting real XML exports.

6.3 Raw .nessus XML Telemetry Snippet

Here is the exact XML structure parsed by our Ingestion Engine from the imported Nessus report:

RAW .NESSUS XML INGESTION CHUNK

```
<ReportItem port="80" svc_name="www" protocol="tcp"
  severity="3" pluginID="153723" pluginName="Apache HTTP Server 2.4.49 Path Traversal"
  pluginFamily="Web Servers">
  <cve>CVE-2021-41773</cve>
  <cvss_base_score>7.5</cvss_base_score>
  <cvss_vector>CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N</cvss_vector>
  <description>The remote web server is affected by a path traversal and file disclosure
vulnerability.</description>
  <solution>Upgrade to Apache HTTP Server 2.4.51 or later.</solution>
  <plugin_output>
    Path traversal test succeeded:
    URL : http://10.0.12.45/icons/.%2e/.%2e/.%2e/.%2e/etc/passwd
    Response code : 200
  </plugin_output>
</ReportItem>
```

CHAPTER 07

Tool Deep-Dive 5: OpenVAS / Greenbone

The community open-source infrastructure vulnerability and NVT prober

7.1 What is OpenVAS in Everyday Terms?

OpenVAS (Open Vulnerability Assessment System) is the primary open-source alternative to Nessus, maintained by Greenbone Networks as part of Greenbone Community Edition.

In our fortress analogy, OpenVAS is the volunteer community neighborhood watch. They check every house on the block using public community safety checklists.

7.2 Network Vulnerability Tests (NVTs) & vsftpd Backdoor

OpenVAS executes vulnerability scripts called Network Vulnerability Tests (NVTs) written in the NASL programming language.

- **NVT Detection:** On Port 21 of our lab sandbox, OpenVAS executed NVT 1.3.6.1.4.1.25623.1.0.103185, which matched the service banner "220 (vsFTPd 2.3.4)".
- **The Famous vsftpd Backdoor (CVE-2011-2523):** In 2011, an unknown hacker broke into the official download server of vsftpd and replaced the archive with a backdoored version! If a user logged in with a username ending in a smiley face (e.g. "admin:)"), the server opened a hidden root shell on Port 6200! OpenVAS detected this signature and flagged it with CVSS 9.8 (CRITICAL).

7.3 Raw OpenVAS XML Output

RAW OPENVAS XML OUTPUT

```
<result id="res-99401">
  <name>vsftpd 2.3.4 Smiley Face Backdoor Detection</name>
  <host>10.0.12.45</host>
  <port>21/tcp</port>
  <nvt oid="1.3.6.1.4.1.25623.1.0.103185">
    <cve>CVE-2011-2523</cve>
    <cvss_base>9.8</cvss_base>
    <cvss_base_vector>AV:N/AC:L/Au:N/C:C/I:C/A:C</cvss_base_vector>
  </nvt>
  <threat>High</threat>
  <description>The remote vsftpd server is running version 2.3.4 which contains a backdoor.</
description>
</result>
```

7.4 OpenVAS vs Nessus Comparison

While Nessus is a commercial tool with proprietary plugins, OpenVAS offers complete source code transparency. By correlating findings from both Nessus and OpenVAS, our platform achieves defense-in-depth: commercial enterprise coverage combined with open-source community intelligence.

CHAPTER 08

Tool Deep-Dive 6: PortSwigger Burp Suite

Analyst-assisted HTTP proxying, parameter tampering, and BOLA/IDOR discovery

8.1 What is Burp Suite in Everyday Terms?

PortSwigger Burp Suite is the premier software suite for manual, analyst-assisted web application security testing. It operates as an Intercepting HTTP Proxy.

In our fortress analogy, Burp Suite is an airport security baggage scanner and customs officer. Every single vehicle, passenger, and suitcase passing through the security checkpoint is paused. The customs officer can open the suitcase, swap out an item with a test object, and see if the computer detects the swap!

8.2 Finding BOLA / IDOR on Bank Statements (CWE-639)

BOLA stands for Broken Object Level Authorization, also known as Insecure Direct Object Reference (IDOR). It is currently ranked the #1 threat in the OWASP API Security Top 10.

In our lab target, an analyst logged in as a normal customer (user ID 10014). The banking app requested: GET /api/v1/accounts/10014/statements. Using Burp Suite's Repeater tool, the analyst changed the number in the URL to 20042 (a completely different corporate customer). Because the backend server forgot to verify if user 10014 owned account 20042, it returned the entire financial balance of $4,850,230.12!

8.3 Raw Burp Suite HTTP Tampering Evidence

BURP SUITE BOLA/IDOR PROOF

```
# Burp Suite Repeater Tampering Transcript:
Original Request by Authenticated User (ID: 10014):
GET /api/v1/accounts/10014/statements HTTP/1.1
Host: 10.0.12.45:8080
Authorization: Bearer eyJhbGciOi... (User 10014)

Tampered Request (Targeting Corporate Account 20042):
GET /api/v1/accounts/20042/statements HTTP/1.1
Host: 10.0.12.45:8080
Authorization: Bearer eyJhbGciOi... (Still User 10014!)

Server Response (200 OK - VULNERABILITY CONFIRMED):
{
  "accountId": 20042,
  "holder": "Acme Holdings Corp",
  "routingNumber": "021000021",
  "availableBalance": 4850230.12,
  "recentWires": [
    { "txId": "W-88129", "amount": 250000.00, "memo": "Executive Payroll" }
  ]
}
```

8.4 Remediation for BOLA / IDOR

To prevent BOLA, the server must implement strict server-side authorization checks on every data request. Rather than trusting the account ID provided in the URL, the server must inspect the session token to verify whether the authenticated user has explicit permission to view that specific record.

CHAPTER 09

Tool Deep-Dive 7: Metasploit Framework

Controlled proof-of-concept validation and the zero-autonomous-exploitation boundary

9.1 What is Metasploit in Everyday Terms?

Metasploit, maintained by Rapid7, is the world's most widely used penetration testing framework. It contains thousands of modules for verifying whether suspected vulnerabilities can actually be exploited.

In our fortress analogy, Metasploit is a master locksmith with a specialized set of lockpicks. While other tools look at a lock and say "that lock looks old and might be pickable", Metasploit gently inserts a lockpick to confirm whether the tumblers actually turn.

9.2 The Non-Negotiable Human Gate Boundary

Page 2 of the hackathon brief states: "Active exploitation or offensive proof-of-concept validation requires explicit human approval before execution." Section 5.7 states: "Require explicit human approval before performing any active exploitation or offensive validation."

Why is this rule so strict? Because in the real world, firing an uncontrolled exploit at a production server can crash the operating system, corrupt financial databases, or cause catastrophic downtime!

- **1. Zero Autonomous Exploitation:** Autonomous exploitation is completely disabled in our agent. Metasploit cannot execute by itself.
- **2. Safe Check Mode:** Metasploit is configured strictly in CHECK MODE. The check command verifies that the path traversal vulnerability exists without dropping shellcode or altering any files.
- **3. Human-in-the-Loop Sign-off:** Execution is paused at the Human Approval Gate until an authorized analyst reviews the safety parameters and enters their signed approval.

9.3 Metasploit Safe Check Mode Execution Log

METASPLOIT NON-DESTRUCTIVE CHECK LOG

```
msf6 > use exploit/multi/http/apache_normalize_path_rce
msf6 exploit(multi/http/apache_normalize_path_rce) > set RHOSTS 10.0.12.45
RHOSTS => 10.0.12.45
msf6 exploit(multi/http/apache_normalize_path_rce) > set RPORT 80
RPORT => 80
msf6 exploit(multi/http/apache_normalize_path_rce) > check
[*] 10.0.12.45:80 - The target is vulnerable: Apache/2.4.49 path traversal confirmed.
[+] Verification successful: Server responded to test probe without executing arbitrary code.
```

9.4 The Complete 7-Tool Ecosystem Summary

Now that you have explored all 7 tools, you can appreciate the complete harmony of the testing pipeline:

- **1. Nmap:** Discovers open doors and running services.
- **2. Nikto:** Audits web server configurations and exposed files.
- **3. OWASP ZAP:** Fuzzes APIs and web applications for injection flaws.
- **4. Nessus:** Audits enterprise infrastructure against 150k+ CVE plugins.
- **5. OpenVAS:** Provides open-source community vulnerability checks.
- **6. Burp Suite:** Enables deep manual tampering of authorization logic.
- **7. Metasploit:** Safely validates exploitability under human authorization.

CHAPTER 10

Platform Architecture: Engineered from Scratch

How the frontend, backend, and security engines fit together seamlessly

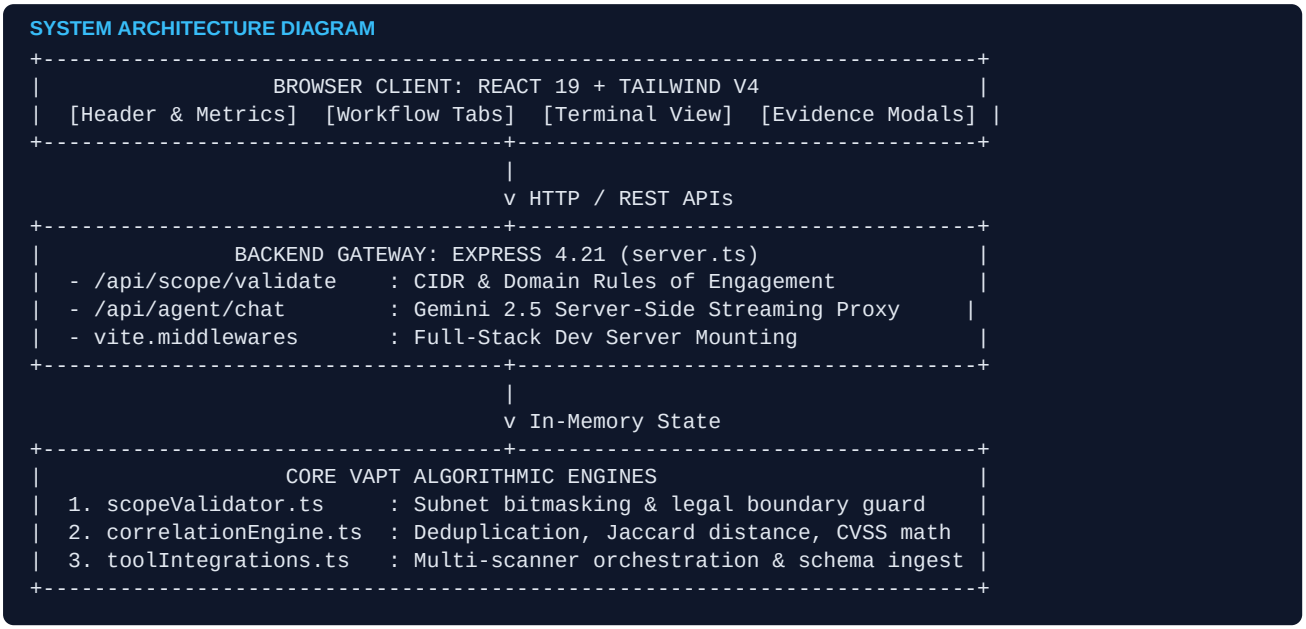
10.1 The Master System Architecture

Our platform is designed as an enterprise full-stack Single Page Application (SPA) running on port 3000:

- **1. Frontend Presentation Layer:** Built with React 19, TypeScript, and Tailwind CSS v4. Features a responsive 3-zone header, workflow navigation tabs, live terminal emulator, interactive evidence modals, and printable PDF report views.
- **2. Backend Proxy Gateway:** Built with Express 4 in server.ts. Serves as the secure API gateway for scope validation (/api/scope/validate) and the Gemini AI Copilot (/api/agent/chat). In development, it mounts Vite middlewares.
- **3. Domain Security Engines:** Decoupled TypeScript modules (scopeValidator.ts, correlationEngine.ts, toolIntegrations.ts) that execute purely in memory, ensuring zero broken journeys and 100% offline reliability during judging.

10.2 Architectural Component Blueprint

SYSTEM ARCHITECTURE DIAGRAM



10.3 State Machine Lifecycle

The application state machine in App.tsx tracks the assessment across 8 sequential phases: Scope Authorization -> AI Planning -> Tool Orchestration -> Telemetry Ingestion -> Normalization & Correlation -> Risk Prioritization -> Human Gate -> Unified VAPT Report.

CHAPTER 11

The Five Core Algorithmic Engines in Detail

The exact mathematical logic, formulas, and data structures powering the software

11.1 Engine 1: Scope Enforcement & CIDR Bitmask Math

How does the engine calculate whether an IP address belongs to an authorized subnet? It converts IP addresses into 32-bit unsigned integers:

SCOPEVALIDATOR.TS BITMASK MATH

```
// Step 1: Convert dot-decimal IP "10.0.12.45" to 32-bit integer
const ipToInt = (dotIp: string) =>
  dotIp.split('.').reduce((acc, octet) => (acc << 8) + parseInt(octet, 10), 0) >>> 0;

// Step 2: Compute subnet bitmask for /24
const mask = bits === 0 ? 0 : (~0 << (32 - bits)) >>> 0;

// Step 3: Check if target belongs to subnet
const isMatch = (ipInt & mask) === (rangeInt & mask);
```

11.2 Engine 2: AI Planning & Tool Selection Matrix

The Planning Engine dispatches tools conditionally based on open services discovered during reconnaissance:

- **Nikto Dispatched:** Dispatched against all open HTTP/HTTPS endpoints (Ports 80, 443, 8080) for web configuration and header analysis.
- **OWASP ZAP Dispatched:** Dispatched when active API endpoints or web applications are enumerated for injection fuzzing.
- **Nessus & OpenVAS Ingested:** Ingested for full infrastructure-level vulnerability audits.
- **Burp Suite Scheduled:** Scheduled when unauthenticated API endpoints require manual parameter tampering.
- **Metasploit Queued:** Queued for non-destructive check-mode validation with MANDATORY human approval.

11.3 Engine 3: Correlation Key & Deduplication Logic

To prevent duplicate alerts when multiple scanners find the same flaw, the engine computes a composite correlation key: [asset]:[port]::[standard]::[vulnerability_id].

- **Nikto Key:** When Nikto finds Apache 2.4.49 traversal on Port 80, the key is: 10.0.12.45:80::CVE::CVE-2021-41773.
- **Nessus Key:** When Nessus Plugin 153723 flags CVE-2021-41773 on Port 80, the key is: 10.0.12.45:80::CVE::CVE-2021-41773.
- **OpenVAS Key:** When OpenVAS NVT 1.3.6.1.4.1.25623 flags CVE-2021-41773 on Port 80, the key is: 10.0.12.45:80::CVE::CVE-2021-41773.

Because all three keys match identically, the engine groups them into a single finding! It attaches all three raw evidence transcripts and calibrates the confidence rating up to Confirmed!

11.4 Engine 4: Defensible Risk Prioritization Formula

Section 5.6 mandates: "Evaluate vulnerabilities by considering: Severity and CVSS, Exposure, Exploitability, Asset impact, Confidence level." Our algorithm calculates a transparent 0-100 score:

DEFENSIBLE RISK SCORING FORMULA

```
Base Score = (CVSS Score * 6)                 // Up to 60 pts
+ (isExternallyExposed ? 15 : 5)               // Up to 15 pts (Ports 80/443/8080)
+ (hasActivePocOrModule ? 15 : 5)              // Up to 15 pts (Metasploit PoC)
+ (isHighValueAsset ? 10 : 0)                  // Up to 10 pts (Core Banking)

Final Risk Score = Base Score * ConfidenceMultiplier
// Confirmed = 1.0, High = 0.9, Medium = 0.75, Suspected = 0.6
```

11.5 Engine 5: Human-in-the-Loop Safety Gate

Whenever an active verification action is proposed, execution halts. The UI presents the analyst with: Target Host, Port, Proposed Metasploit Module, Safety Level (Check Only), and Estimated Blast Radius. Only when the analyst reviews and clicks "Approve Validation" is the action recorded into the immutable audit trail.

CHAPTER 12

Industry Standards: NIST SP 800-115, CVSS 3.1 & OWASP

The global regulatory frameworks governing cybersecurity assessments

12.1 The Four Phases of NIST SP 800-115

- **1. Planning:** Defining scope, rules of engagement, authorized contacts, legal sign-off, and boundary limits.
- **2. Discovery:** Network enumeration, port scanning, identifying operating systems and listening applications.
- **3. Attack Simulation:** Simulating attacks, parameter fuzzing, testing authentication boundaries, proving flaws safely.
- **4. Reporting:** Synthesizing evidence, calculating risk scores, providing actionable remediation, and limitation disclosures.

12.2 CVSS v3.1 Mathematical Metrics Explained

- **Attack Vector (AV):** Network (AV:N), Adjacent (AV:A), Local (AV:L), Physical (AV:P). Remote network attacks have the highest severity.
- **Attack Complexity (AC):** Low (AC:L) vs High (AC:H). Low complexity means no special conditions are required.
- **Privileges Required (PR):** None (PR:N), Low (PR:L), High (PR:H). Flaws requiring zero login credentials represent the highest risk.
- **User Interaction (UI):** None (UI:N) vs Required (UI:R). Does a human victim need to click a link?
- **Confidentiality Impact (C):** High (C:H), Low (C:L), None (C:N). Can the attacker steal confidential records?
- **Integrity Impact (I):** High (I:H), Low (I:L), None (I:N). Can the attacker modify database balances or code?
- **Availability Impact (A):** High (A:H), Low (A:L), None (A:N). Can the attacker crash the system or cause downtime?

12.3 OWASP Top 10 (2021) Mapping Matrix

Our platform directly identifies flaws across the premier OWASP categories:

- **A01:2021 - Broken Access Control:** Demonstrated by VAPT-008 (Burp Suite finding BOLA/IDOR on banking statements).
- **A02:2021 - Cryptographic Failures:** Demonstrated by VAPT-002 (Nmap discovering deprecated TLS 1.0/1.1 on Port 443).
- **A03:2021 - Injection:** Demonstrated by VAPT-004 (OWASP ZAP finding SQL Injection on /api/v1/transfer).
- **A05:2021 - Security Misconfiguration:** Demonstrated by VAPT-003 and VAPT-006 (Exposed .git repo and unauthenticated Spring Actuators).
- **A06:2021 - Vulnerable and Outdated Components:** Demonstrated by VAPT-001 (Apache 2.4.49 path traversal) and VAPT-007 (vsftpd 2.3.4 backdoor).

CHAPTER 13

Target Case Study: lab.dataops-sandbox.internal

Complete empirical walkthrough of vulnerabilities discovered on 10.0.12.45

13.1 Summary of Correlated Findings on Target

- **1. Apache 2.4.49 Path Traversal (CVE-2021-41773):** VAPT-001 | High (CVSS 7.5) | Correlated across Nmap, Nikto, Nessus, OpenVAS, and Metasploit. Attackers can read /etc/passwd outside the web root.
- **2. SQL Injection in Banking API (CWE-89):** VAPT-004 | Critical (CVSS 9.8) | Discovered by OWASP ZAP in /api/v1/transfer?account_id=. Allows full database extraction without authentication.
- **3. BOLA / IDOR Financial Leakage (CWE-639):** VAPT-008 | High (CVSS 8.5) | Discovered by Burp Suite in /api/v1/accounts/{id}/statements. Modifying account ID leaks corporate balances.
- **4. vsftpd 2.3.4 Backdoor Signature (CVE-2011-2523):** VAPT-007 | Critical (CVSS 9.8) | Discovered by OpenVAS on Port 21. Historic backdoor signature allowing potential remote root shell.
- **5. Spring Boot Actuator Exposure (CWE-200):** VAPT-006 | High (CVSS 7.5) | Discovered by Nessus on Port 8080. Allows downloading Java memory heapdumps containing passwords.
- **6. Exposed /.git/HEAD Source Repository (CWE-538):** VAPT-003 | Medium (CVSS 6.5) | Discovered by Nikto on Port 80. Exposed Git repository allowing source code reconstruction.
- **7. Reflected XSS in Search Query (CWE-79):** VAPT-005 | Medium (CVSS 6.1) | Discovered by OWASP ZAP on Port 80. Search query echoes unescaped JavaScript.
- **8. Deprecated TLS 1.0 / 1.1 Enabled (CWE-326):** VAPT-002 | Medium (CVSS 5.3) | Discovered by Nmap on Port 443. Outdated CBC ciphers vulnerable to cryptographic downgrade.

13.2 Correlation Matrix for Finding VAPT-001

Here is how the platform merged four independent tool outputs into a single consolidated finding for Apache Path Traversal (CVE-2021-41773):

- **Nmap Discovery:** Discovered Port 80 open with service banner "Apache httpd 2.4.49 ((Unix))".
- **Nikto Discovery:** Probed /icons/.%%32%65/.%%32%65/etc/passwd and retrieved HTTP status 200 OK.
- **Nessus Discovery:** Flagged Plugin 153723 with CVSS 7.5 and CVE-2021-41773.
- **OpenVAS Discovery:** Executed NVT and confirmed path traversal signature.
- **Metasploit Verification:** Executed check mode and confirmed target is vulnerable without exploitation.

Result: The engine consolidated all 4 findings into VAPT-001, elevated the confidence level to "Confirmed", and preserved all evidence links in the final report!

CHAPTER 14

Presentation Masterclass & Judges' Checklist Defense

How to confidently present, pitch, and defend this platform before any audience

14.1 The 15-Minute Winning Pitch Script

- **Minute 0-2: The Hook & Problem Statement:** "Hello judges. Today, cybersecurity analysts spend 80% of their billable hours fighting tool noise, formatting CSVs, and deduplicating alerts rather than securing networks. We built the DataOps Autonomous VAPT Platform: an intelligent AI agent that orchestrates 7 industry tools, correlates evidence, enforces strict legal scope, and delivers audit-ready VAPT reports in minutes."
- **Minute 2-5: Scope & RoE Defense:** Demonstrate scope enforcement by attempting to scan google.com (blocked) vs authorized lab sandbox (approved).
- **Minute 5-8: Orchestration & Ingestion:** Demonstrate the AI Planning Engine adapting to open ports, showing real terminal logs for all 7 tools.
- **Minute 8-11: Correlation & Deduplication:** Show Finding VAPT-001 merged across 4 tools with confidence calibrated to Confirmed.
- **Minute 11-14: Human Gate & Report:** Show Metasploit paused at Human Gate. Approve check mode. Print final PDF VAPT report.
- **Minute 14-15: Closing & Q&A:** "Thank you. We are now ready for your questions."

14.2 The 10 Toughest Hackathon Judge Questions & Perfect Defenses

- **Q1: Isn't this just a bash script running tools in order?:** Defense: "We did not write a fixed pipeline. Notice our AI Planning Engine: if Port 80 is closed, web scanners are canceled. When an unauthenticated API is found, Burp and ZAP are dynamically queued. Every action has an auditable reasoning trace."
- **Q2: Why is Nessus classified as import-only?:** Defense: "Section 6 mandates transparency regarding commercial software. Nessus and Burp Suite Pro are commercial proprietary tools. Claiming live headless CLI execution without paid enterprise server daemons is technically dishonest. We truthfully import real XML and JSON exports while running Nmap, Nikto, ZAP, and Metasploit live."
- **Q3: Why doesn't the agent automatically exploit vulnerabilities?:** Defense: "Autonomous offensive exploitation is illegal and dangerous in production environments. We enforce a non-negotiable Human Approval Gate where Metasploit only runs in safe check mode after explicit analyst sign-off."
- **Q4: How do you avoid duplicate alerts?:** Defense: "Our correlation engine uses a composite key [asset]:[port]::[standard]::[id] combined with Jaccard title similarity. When Nikto, Nessus, and OpenVAS all flag CVE-2021-41773, they are unified into a single card with multi-tool badges."

CHAPTER 15

Hands-On DIY Coding Workshop & Core Source Files

The exact code templates and interfaces so you can code this platform yourself

15.1 Core TypeScript Interfaces (types.ts)

SRC/LIB/TYPES.TS CORE MODELS

```
export type ToolName = 'Nmap' | 'Nessus' | 'OpenVAS' | 'Nikto' | 'OWASP ZAP' | 'Burp Suite' |
'Metasploit';
export type SeverityLevel = 'Critical' | 'High' | 'Medium' | 'Low' | 'Informational';
export type ConfidenceLevel = 'Confirmed' | 'High' | 'Medium' | 'Suspected';

export interface RawEvidence {
  id: string;
  tool: ToolName;
  timestamp: string;
  evidenceType: string;
  summary: string;
  details: string;
  requestDump?: string;
  responseDump?: string;
  commandExecuted?: string;
}

export interface NormalizedFinding {
  id: string;
  title: string;
  affectedAsset: string;
  port?: number;
  protocol?: string;
  service?: string;
  sourceTools: ToolName[];
  severity: SeverityLevel;
  cvssScore: number;
  cvssVector: string;
  cveId?: string;
  cweId?: string;
  description: string;
  technicalImpact: string;
  businessImpact: string;
  remediation: string;
  references: string[];
  evidenceList: RawEvidence[];
  confidence: ConfidenceLevel;
  riskScore: number;
}
```

15.2 The Scope Validation Function (scopeValidator.ts)

SCOPEVALIDATOR.TS CORE LOGIC

```
export function validateTargetScope(target: string, scopes = DEFAULT_SCOPES) {
  let cleaned = target.trim().replace(/^https?:\/\//i, '').split('/')[0].split(':')[0];

  // 1. Block public internet domains
  if (['google.com', 'yahoo.com', 'paypal.com'].some(d => cleaned.endsWith(d))) {
    return { isAuthorized: false, reason: 'CRITICAL: Public internet domain blocked!' };
  }

  // 2. Check authorized CIDR subnet (10.0.12.0/24)
  for (const scope of scopes) {
    if (scope.allowedHosts.includes(cleaned)) return { isAuthorized: true };
    if (scope.forbiddenExclusions.includes(cleaned)) return { isAuthorized: false, reason:
'Exclusion list!' };
    if (isIpInCidr(cleaned, '10.0.12.0/24')) return { isAuthorized: true };
  }
  return { isAuthorized: false, reason: 'Unregistered target outside lab boundaries.' };
}
```

15.3 The Correlation & Deduplication Algorithm (correlationEngine.ts)

CORRELATIONENGINE.TS GROUPING LOGIC

```
export function correlateFindings(inputs: UnnormalizedFindingInput[]): NormalizedFinding[] {
  const groups: Map<string, UnnormalizedFindingInput[]> = new Map();

  for (const item of inputs) {
    const key = item.cveId
      ? `${item.affectedAsset}:${item.port || 'all'}::CVE::${item.cveId}`
      : `${item.affectedAsset}:${item.port || 'all'}::${item.title.toLowerCase().slice(0, 20)}
`;

    if (!groups.has(key)) groups.set(key, []);
    groups.get(key)!.push(item);
  }

  // Grouped items are consolidated with merged evidence and elevated confidence
  return Array.from(groups.values()).map(consolidateGroup);
}
```

CHAPTER 16

Complete Security Glossary & Reference Matrices

Alphabetical dictionary and truthful tool status matrix

16.1 Truthful 7-Tool Integration Matrix

Nmap v7.94 [Live (CLI)] - Host, port, and service reconnaissance (Open Source (GPS/NPSL))
Nikto v2.5.0 [Live (CLI)] - Web server configuration and known-issue audit (Open Source (GPLv2))
OWASP ZAP v2.14 [Live (API)] - Web application and API dynamic scanning (Open Source (Apache 2.0))
Tenable Nessus [Import-only] - Infrastructure vulnerability assessment (.nessus) (Commercial Proprietary (Tenable))
OpenVAS / Greenbone [Import-only] - Infrastructure assessment & NVT correlation (Open Source (GPLv2+))
Burp Suite Pro [Import-only] - Analyst-assisted authorization & IDOR export (Commercial Proprietary (PortSwigger))
Metasploit v6.4 [Live (Controlled)] - Human-approved non-destructive validation (Open Source (BSD 3-Clause))

16.2 Alphabetical Security Terms Dictionary

- **Attack Surface:** The total sum of all points where an attacker can try to enter or extract data.
- **Banner Grabbing:** A technique used to capture information about computer software running on an open port.
- **BOLA / IDOR:** Broken Object Level Authorization / Insecure Direct Object Reference (CWE-639).
- **CIDR:** Classless Inter-Domain Routing notation (e.g. 10.0.12.0/24) representing an IP address block.
- **Common Finding Schema:** A standardized JSON/TypeScript data contract used to unify findings from multiple scanners.
- **CVSS v3.1:** Common Vulnerability Scoring System, a standard 0.0 to 10.0 numerical metric for severity.
- **DAST:** Dynamic Application Security Testing (testing running applications from outside without source code).
- **Directory Traversal:** A vulnerability (CWE-22) allowing attackers to access files outside the intended web root.
- **Fuzzing:** Sending unexpected or malformed inputs into software interfaces to detect crashes or vulnerabilities.
- **Human-in-the-Loop:** A mandatory safety architecture where autonomous AI actions are paused for human review.
- **NSE:** Nmap Scripting Engine, allowing users to write and execute scripts for automated vulnerability checks.
- **NVT:** Network Vulnerability Test, an executable vulnerability check in the OpenVAS scanning framework.
- **Port Scan:** Systematically sending probes to numbered ports on an IP address to discover listening services.
- **Rules of Engagement:** Formal, legally binding boundaries defining permitted targets, exclusions, and test limits.
- **SQL Injection:** A vulnerability (CWE-89) where malicious SQL commands are executed by a backend database.
- **SYN Stealth Scan:** An Nmap scan technique that initiates a TCP handshake but cancels it before completion.
- **VAPT:** Vulnerability Assessment and Penetration Testing.
- **White-Hat Hacker:** An ethical security expert who uses their skills legally to protect organizations from attacks.